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Detection and Identification of Dyes on pre–Hadrianic Textiles from Vindolanda

G. W. TAYLOR

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INTRODUCTION AND OBJECTIVES

A considerable collection of textile fragments was discovered at Vindolanda during the excavations of 1972–1975, amongst the debris from the wooden military buildings which underlie the civilian houses in the south-eastern corner of the *vicus*. The unusual local conditions — anaerobic but not waterlogged — contributed to the preservation of the fragments. All the textiles came from the levels associated with the two earliest occupations of the site, which fall within the period AD 85–115, and probably closer to 90–105. No textiles were found in the later wooden levels (dating AD 115 to *c.* 125), but there the soil conditions were considerably less kind to organic material, and it is unlikely that textiles would have survived.

All the textiles were of sheep's wool, and most have been described by Wild and Ryder.[1] They were all rags, and many showed signs of considerable wear. The view was that they were produced in northern Britain, from local wool, although final proof is lacking. Over 50 yarns from the textiles were examined by Ryder.[2] Only 9% were completely coloured (that is, heavily pigmented), and 40% were white. However, 51% had a mixture of pigmented and white fibres, indicating grey fleeces. The white and lightly coloured wools would have been suitable for dyeing with bright colours.

The survival of so many textile fragments, many made with unpigmented wool, presented an opportunity to gain some direct information on dyes and dyeing techniques that may have been used on the northern frontier of Roman Britain only 50 years after the Claudian invasion. All the textiles were heavily stained from burial and there was little or no visual evidence for dyes. However, the textiles had only been rinsed in water in the conservation process, so any residual dye had a good chance of remaining intact. The chances of dye surviving the processes of loss and decay due to burial for such a long time seemed small; but any information from an experimental study would shed light on an aspect of British textile technology of which hitherto we had little or no direct evidence.

BACKGROUND TO THE ROMANO-BRITISH DYEING ACTIVITY

By the first century, sophisticated dyes and dyeing procedures were well established in southern Europe and the Middle East. A number of dyes had widespread use. The standard blue dye was indigotin, almost entirely from the plant woad. This was the heyday of the shellfish purple, Tyrian Purple, whose use in the Roman Empire was restricted at this time to emperors and certain high-ranking officers. The colorant is a close relative of indigotin. Purples were also made from blue and red mixtures, and from the purple lichen dye, orchil; the latter

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is fugitive and would not be expected to survive. The standard red was common madder (*Rubia tinctorum* L.) which the Romans were cultivating. A prestigious red was the insect dye, kermes, and another red, similar to kermes and modern cochineal, was also known.[3] A good yellow was available from the plant weld, and chemically related yellow dyes were found in other common plants. Many dyes enjoyed local, if not general, use and one might expect the use of local dyes to become more prevalent as one moves from the centre of the Roman world north-west into Britain.

Whether or not the Iron Age Britons used bright colours is clouded by a number of factors. First, the main British Iron Age sheep had a brown fleece,[4] but the Vindolanda textiles themselves constitute a principal piece of evidence that unpigmented wool was also available at this time. To make dyeing worthwhile, this white wool would have had to have been reasonably freely available; though of course the Vindolanda textiles might have been special as part of fulfilling orders for the Roman army. Second, a high proportion of dyes giving bright and permanent colours were mordant dyes, requiring a mordant such as alum. An essentially iron-free mineral is required and, at this time, only material not requiring processing was used. This was not available in Britain and, therefore, if alum was used by the Romano-British, it was surely imported material. Third, although a range of local dyes was available in Northern England in the first century, the main commercial red, common madder, was not produced locally. This plant is not indigenous to Britain, and there is no evidence that it was ever grown in the north. Madder is a member of the Galieae tribe, which also contains a number of plants which could, in principle, have been used instead of madder. All, however, require a mordant such as alum. Indigotin may have been available from woad, but, again, this is not a wild plant and its cultivation would have been necessary. Weld, which is now common over much of Britain, was probably available as a good yellow (with alum mordant). There is a tradition of lichen dyes in Scotland and Scandinavia, and quite possibly such dyes were used at Vindolanda; the Vindolanda Trust have shown that local lichens are capable of giving good colours on wool.

The Roman army tended to rely on local supplies of cloth, and many of the textile fragments found at Vindolanda were probably used by the garrison of the fort. Though the fabrics were probably woven as a home industry, little is known about the organization of the finishing activity, or how dyeing was done. In later Roman times, documents refer to a *gynaeceum* (a centralized fabric-making establishment) in Britain,[5] and this may have included a dye works.[6] However, it does seem likely that any dyeing in first-century Vindolanda was a domestic activity only.

Red Dyes from Plants of the Galieae Tribe

Plants of the Galieae tribe have been used for millennia as sources of dyes which, with alum mordant, give red colours on wool. The common colorant of all the members of the group is pseudopurpurin, a hydroxyanthraquinone.[7] Many of the plants also contain another hydroxyanthraquinone, alizarin, which is regarded as the main colorant in common madder. Purpurin, a close relative of alizarin, is also found in common madder, and also in wild madder (*R. peregrina* L.), which is deficient in alizarin. Also present in the madders are lesser quantities of other hydroxyanthraquinones; indeed, 19 such compounds have been

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found in common madder,[8] most of these contributing to the colour of the dye.

Neither of the madders would be freely available at Vindolanda. The wild variety does indeed still grow in Britain, but only on the south and west coasts.

The bedstraws belong to the *Galium* species and are similar to the madders in their colorants.[9] Several bedstraws were probably available in first-century Vindolanda, and all contain pseudopurpurin, alizarin and purpurin.

At first sight, therefore, it appears difficult to differentiate between madder and bedstraw on the basis of analysis of colorant on dyed fibres. However, in the present study, analysis of modern alum-mordanted wool dyed with common madder and with Lady's Bedstraw shows a major difference. Whereas alizarin is prominent in the madder dyeing, it is absent in the bedstraw dyeing. This finding is in accord with the early work of Hill and Richter[10] who claim that the bedstraw contains negligible alizarin. The difference between the results of Hill and Richter (and the analysis of bedstraw-dyed wool) on the one hand and the exhaustive analytical search by Burnett and Thomson[11] on the other may lie in the availability of components to extraction. On the present evidence, a bedstraw-dyed wool is less likely to contain alizarin than a madder-dyed wool.

Wild madder is similar to Lady's Bedstraw in being deficient in (available) alizarin. Indeed, it is difficult to distinguish analytically between dyeings from the two plants.

EXPERIMENTAL PROCEDURE

The procedures for the extraction and identification of dyes have been described elsewhere.[12] In outline, indigotin and certain lichen dyes are extracted with hot pyridine/water (1/1 by volume), which also extracts some of the 'background dirt'.[13] The dyes are separated from the diluted pyridine/water mixture by extraction with ether. The same textile sample is then extracted with ethanol/10% aqueous sulphuric acid (2/1 by volume). The acid hydrolyses the mordant–dye link and the alcohol extracts the dye. The alcohol is evaporated and the dye extracted from the residual liquor with ether. The ether is evaporated and the dye residue taken up in methanol. The absorption spectra of both extracts can then be measured. In the case of the mordant dye extract, it is preferable to add a little crystalline magnesium acetate to the solution. With many mordant dyes, this compound causes the spectrum of the dye to be shifted to longer wavelengths, minimizing the overlap of the dye spectrum with that of the 'background dirt'. In many cases, also, the dye spectrum in the presence of magnesium salt is a useful fingerprint for identifying the dye. (The reaction with magnesium acetate does not occur in ether).

In this work, the spectra were measured on a Perkin–Elmer model 402 recording spectrophotometer over the wavelength range 390–750 nm, using glass microcells. The spectra were then compared with standards, generally obtained by extracting modern wool samples dyed with known dyes. In most cases, at least two tests were carried out, if sufficient sample were available; about 10 mg of fibre were used per test, although much less is required with visibly dyed samples.

Most yellow dyes would not be detected by this procedure, partly because of strong overlap between the spectra of dye and dirt. A number of lichen dyes are not extracted by the solvents used, and possibly certain other potential dyes might not be extracted. Identification of the mordants used has not proved practical since the textiles, having

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been buried for nearly 2000 years, are likely to be contaminated with metals such as iron or aluminium whether mordanted or not.

The composition of certain dye extracts, for example those from madder and Lady's Bedstraw dyeings, was further investigated by paper chromatography, using the solvent system, ethanol/water/ether (4/4/1 by volume) and developing the chromatograms by spraying with methanolic potassium hydroxide solution. Alizarin and purpurin produce blue or purple spots, pseudopurpurin a rose streak, and xanthopurpurin and munjistin, orange/pink spots. This solvent system also resolves some yellow dye mixtures.

Results

Eighty-nine samples, from 53 textiles (including two felts) were tested. Nine samples, also representing nine fabrics, contained definite traces of dye which, in general, could be identified, and these results are described as follows.

The data on the nine dyed textiles are shown in Table 1.

Samples 14/11, 35/9, and 43/46: Madder

At least two of these textiles were made from unpigmented wool as one might expect for a dyed textile. The dye is identified as common madder. The spectra are compared

Table 1. Results on Dyed Textiles

Textile Cat./ inv.[a]	Description[b]	Pigmented wool?[c]	Dye tests: Number done	Dye tests: Number positive[d]	Identification of dye
4/14	Hairy plain weave	Some	5	3	Alizarin
10/53	Simple check, with fringe	Some?	5	1 + 2 poss.	'Purple'
14/11	Fine half basket weave	Almost none	2	2	Madder
17/34a	Tapestry woven band	No?	3	3	Pseudopurpurin
31/7	Medium fine diamond twill	No	4	1 + 1 poss.	Alizarin
35/9	Medium fine diamond twill	No	2	1	Prob. madder
40/41	Diamond twill	?	4	1 + 1 poss.	Purpurin
43/46	Fine diamond twill with hem	?	2	1	Prob. madder
44/86	Fine diamond twill	?	3	3	Pseudopurpurin

[a] Textiles identified by catalogue/inventory numbers as shown in Wild (1977), see Note 1.
[b] Descriptions from Wild (1977).
[c] ? indicates not published in Wild (1977).
[d] A possibly positive test is a result which, by itself, is not definitive, but supports a separate definitely positive result: for example, a very weak spectrum agreeing with a more definite spectrum, or (in the case of 10/53) a purple colour in the second test solution, which disappeared in the later stages of the test.

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with that of madder in Fig. 1. The peak at 525 nm, together with contributions to the shoulders at 500 and 550 nm, is due to pseudopurpurin. The shoulder at about 600 nm is mainly due to alizarin.

The spectrum of madder is similar to that of the dye from Lady's Bedstraw, which is shown in Fig. 4. The absorption is mainly due to pseudopurpurin. With textiles containing little residual madder, the spectrum may be too weak to unequivocally show alizarin absorption. In this case, the dye extract should be paper chromatographed, as described in the experimental section. This was done with sample 14/11, but was not possible with the other two. However, the spectra from 35/9 and 43/46 are more in accord with that of common madder than with that of the bedstraw.

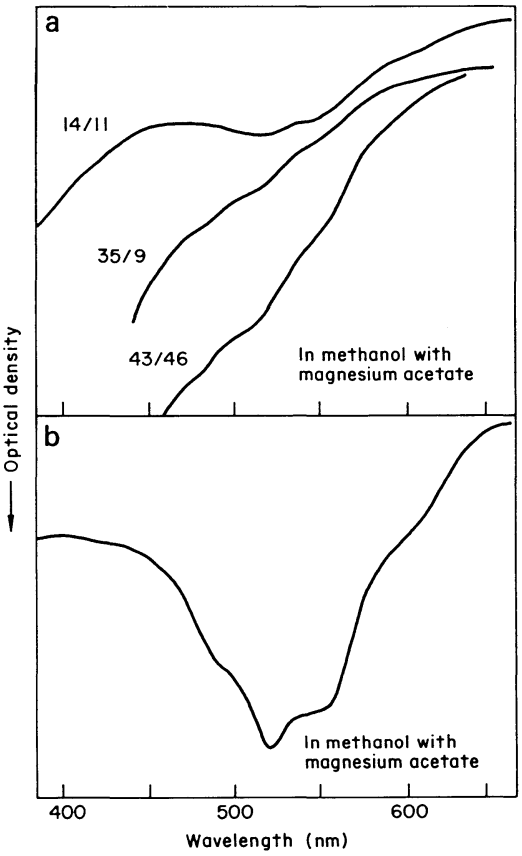

FIG. 1. (a) Dye from textiles 14/11, 35/9 and 43/46; (b) madder.

Samples 10/53: 'Purple'

This was the largest piece of a textile in 11 fragments; none of the other fragments showed evidence for dye. At least some of the wool in the textile, which had a check pattern, was pigmented, and this fragment had traces of a fringe.

The dye tests gave remarkable results. During the first test, the pyridine/water extraction liquor quickly went purple, and the colorant survived to have its spectrum measured. In the second test, the liquor again went purple, but the colour faded in the course of the test. In the third test, the liquor never showed the purple colour, but the spectrum gave a slight suggestion of the purple component. Unfortunately, it is not possible to say whether the positive test results were associated with the testing of the fringe only or of the body of the textile only.

The colour of the test solution was very sensitive to pH (the solution in methanol of the ether extract of the raw pyridine/water extract). When the solution was acidified with sulphuric acid, it turned pink, and with potassium hydroxide, it turned blue. This behaviour is, of course, very similar to that of the common indicator litmus. The corresponding spectrum changes are shown in Fig. 2.

The Roman purple dye, orchil, was produced by fermentation (with stale urine and lime) of lichens of the *Rocella* group, particularly *Rocella tinctoria* D.C.[14] Very similar dyes are still made from lichens such as *Ochrolechia tartarea* (L.) Mass., and lichens of the *Umbilicaria* group.[15] In these dyes, and also in orchil, the basis of the

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colorant is the same, namely orcein. A derivative of orcein is, indeed, litmus.

Samples of the wool dyed purple with dyes from *Ochrolechia tartarea*, three lichens of the Umbilicaria group, and also a 19th-century Spanish wool sample believed to be dyed with orchil, were subjected to the pyridine/water extraction test. A substantial proportion of the dye from all five samples was extracted, and the spectra from the five samples were to all intents and purposes identical, suggesting that the dye from all five samples was essentially the same. The spectrum of the dye from the *Ochrolechia* lichen is shown in Fig. 2; like the dye in 10/53, it was pH sensitive. The modern lichen dyes are all a rich purple; the orchil-dyed Spanish wool is also purple, but of a somewhat reddish hue.

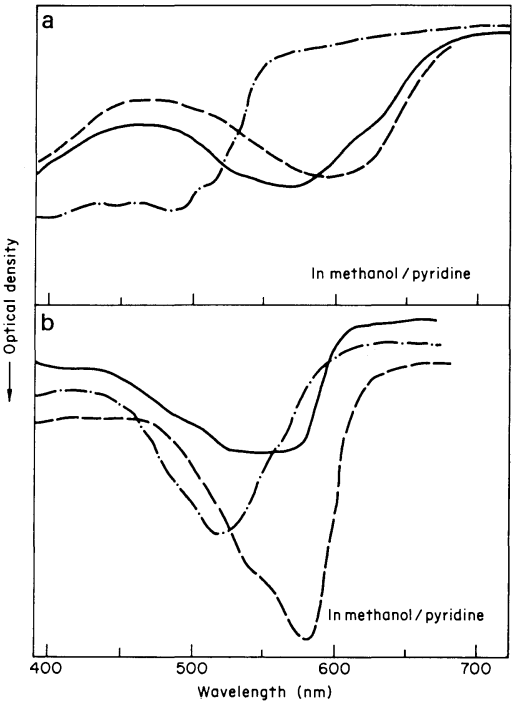

FIG. 2. (a) Dye from textile 10/53; (b) dye from fermented ochrolechia tartarea. ——, as extracted; ---, plus KOH; –·–, plus H_2SO_4.

The spectra from the dye in textile 10/53 are similar, in a general sense, to those of the purple lichen dyes. However, if the 'neutral' spectrum is representative of the original colour of the textile, the absorption at longer wavelengths than the reference dyes suggests that the old dye would have a bluish hue. It is clear that the dye on the textile was in an unstable state (for example, the fading of the dye extract in the second test), and the species actually giving the spectrum may be a derivative of the original dye.

In view of the uncertainty in identifying the dye as orchil, other possible sources for the colour were explored. Berries such as the bilberry, elderberry, and blackberry were used as sources of purple dye in antiquity.[16] However, the spectra of aqueous solutions of the berry colorants did not agree closely with that of the purple in 10/53. Furthermore, dyeings of the berries on wool did not behave similarly to the purple on 10/53 in the pyridine/water extraction test.

It does, therefore, seem more likely that the dye on 10/53 was a purple lichen dye rather than a berry dye. The Vindolanda Trust have indeed produced a purplish dye from a local lichen, *Evernia prunastri* L., which was used in making orchil.

Samples 4/14 and 31/7: Alizarin

The wool in sample 4/14 contained some pigment, whereas that in 31/7 was unpigmented. Comparison of the spectra of the dye from these textiles with that of alizarin (see Fig. 3) suggests that the main colorant is indeed alizarin.

The detection of only alizarin does not necessarily imply that the original dye contained only that colorant. Perhaps selective uptake of alizarin was achieved during dyeing, or possibly other colorants have decayed in the many centuries during

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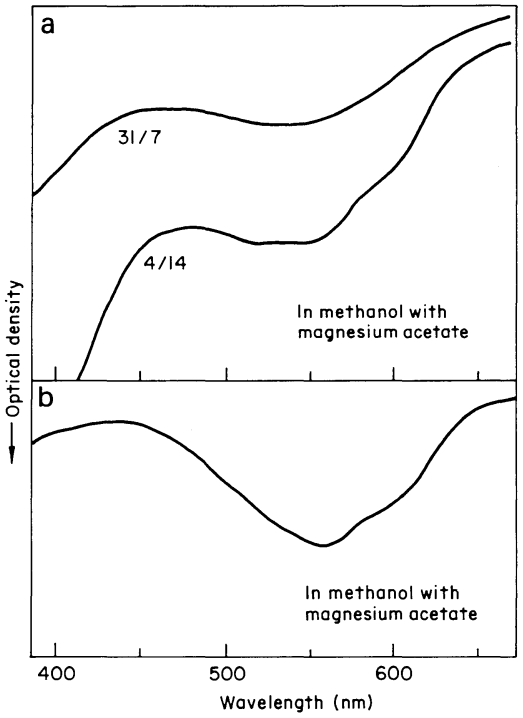

FIG. 3. (a) Dye from textiles 4/14 and 31/7; (b) alizarin.

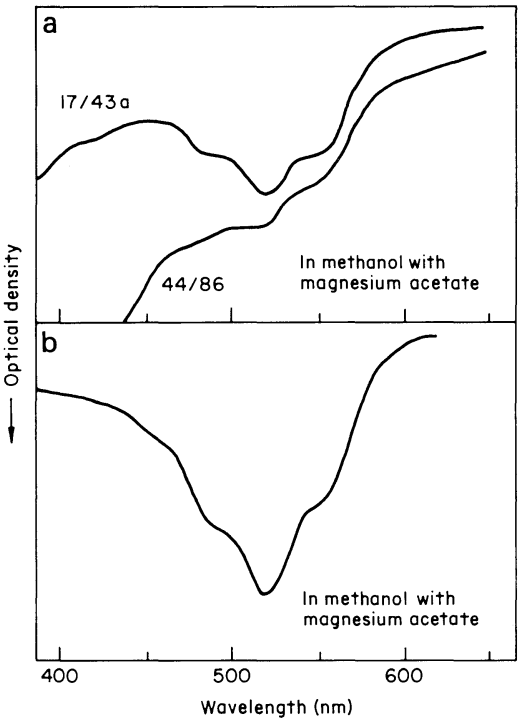

FIG. 4. (a) Dye from textiles 17/43a and 44/86; (b) pseudopurpurin.

which the textile was buried. The most likely source of alizarin is common madder.

Samples 17/43a and 44/86: Pseudopurpurin

Sample 17/43a is a tapestry woven band taken from 17/43, which, when freshly excavated, was thought to have a purplish appearance. It looked like a purple stripe. The sample did indeed prove to be dyed, but with a dye rich in pseudopurpurin, which, with alum mordant, gives a rose colour. The spectrum of the dye is compared with that of pseudopurpurin in Fig. 4. Chromatography of two of the three extracts (the third was not chromatographed) suggested the absence of alizarin. Similarly, chromatography of one of the extracts from 44/86 also suggested the absence of alizarin.

There are several possible explanations. Perhaps the dye used was indeed Lady's Bedstraw, rich in pseudopurpurin but deficient in alizarin. Or perhaps madder was used but the alizarin has been lost.

Sample 40/41: Purpurin

Here the spectrum was a good match for that of purpurin (see Fig. 5) in the single definite test result. The spectrum in the second test, giving a possible indication of purpurin, was not in itself definitive.

Purpurin is a minor tinctorial component in the madders, and is also reported to be present in Lady's Bedstraw.[17] However, the present study suggests that purpurin is at best only a small part of the colorant taken up by the wool. It seems most likely that the

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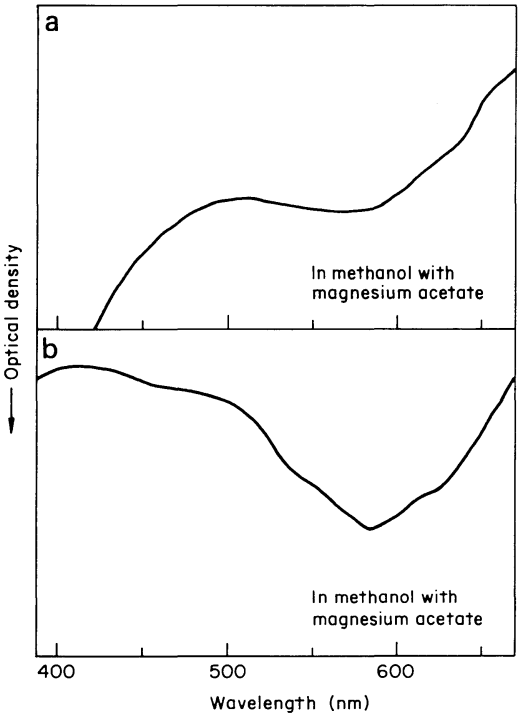

FIG. 5. (a) Dye from textile 40/41; (b) purpurin.

purpurin is a decomposition product of pseudopurpurin, perhaps formed while the textile was buried.

General Features of 'Background' Absorption

In the pyridine/water test, 65 of the 89 samples tested gave spectra with some or all of the following features: peak at about 410 nm, peaks or shoulders at about 595 or 650 nm, and occasionally shoulders at about 480 and 550 nm. The spectrum of the sample giving the most extreme version of these characteristics, textile 59/78, is shown in Fig. 6.

The combination of absorption at 410 and 595 nm tends to give a green colour. It may represent the residues of a green dye (or blue and yellow mixture) — though since nearly three-quarters of all the textiles gave the absorption, it would represent a strong liking for a green colour! It is interesting that, of the nine textiles positively identified as dyed, only one (and in one test only) had the 'background' absorption characteristic — in other words, a red-dyed fabric was not also dyed green! Another possible explanation is that dyed fabrics tended not to absorb green material during burial, while undyed textiles were not so protected.

The material giving the absorption at 410 nm did not show an intensified colour with magnesium acetate or potassium hydroxide, unlike many yellow dyes investigated. The absorption at 595 nm is not due to indigotin, the dye component of woad, which, under these conditions, absorbs at 605 nm. Though the spectrum bears some resemblance to that of chlorophyll, it is not close enough to make the identification likely.

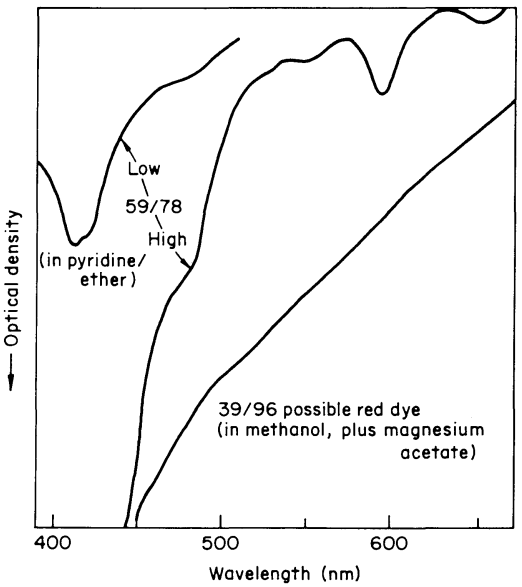

FIG. 6. Extreme 'background' and possible red dye.

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Samples that may have been dyed red

In the test for mordant dyes, the spectra of some of the samples showed (in the presence of magnesium acetate) some signs of absorption at 500–550 nm without giving recognizable bands. The spectrum from textile 39/96, shown in Fig. 6, is typical, and is consistent with the presence of red dye. Sixteen of the 89 samples showed such absorption. However, when the extract from one of these, 27/98, was chromatographed, the chromatograms showed no signs of spots attributable to madder, bedstraw, kermes or cochineal.

DISCUSSION

Nine of the samples tested showed traces of dyes which were generally identifiable. Eight of the samples yielded various hydroxyanthraquinones which are the main colorants in the mordant dye madder. Only three of the textiles appeared to contain all the main components of madder, however. In two of the textiles was present only pseudopurpurin, found in common and wild madders, and also related species such as the bedstraws. Though common madder would probably be an import, the bedstraws were probably growing in the vicinity of Vindolanda at the time.

Two textiles revealed traces of alizarin only. Alizarin is a principal colorant in both common madder and, according to modern research, most of the bedstraws, but only in conjunction with pseudopurpurin. Perhaps the dye actually used was madder or bedstraw, and the other colorants have been lost; or selective uptake of alizarin was achieved in some way at dyeing. The same argument can be advanced in connection with the textiles dyed with pseudopurpurin and the single purpurin-dyed textile, but, in all cases, it is difficult to visualize mechanisms of selective dyeing or selective loss.

One of the fragments from textile 10, 10/53, showed traces of what seems to have been a purple dye. The dye has not been positively identified, but shows some resemblance to lichen purple dyes, including a 19th-century sample of orchil. The dye on 10/53 may be the Roman orchil. The dye may have been partly decomposed, and this may account for the lack of close correspondence between its spectrum and those of the reference purples.

A number of textiles showed some signs of having been dyed with red mordant dye, but the evidence is flimsy. On the other hand, most of the textiles showed signs of extractable green (or rather, yellow and blue) colouring matter. There is some literary evidence for the use of green in civilian clothing but its use by the army is unlikely. It seems plausible to assume that the green coloration is contamination picked up while the textiles were buried.

The picture of dyeing activities at Vindolanda that emerges is a tantalizing one. Since nearly one-fifth of the textiles still retain identifiable traces of dye, it is reasonable to conclude that, originally, a higher proportion of the fragments had been actually dyed — particularly so if one assumes that yellow dyes and lichen dyes, many of which would not have survived or have been detected in this work, were available. The coloured compounds identified require, in general, sophisticated dyeing procedures involving the use of imported materials. If bright colours were to be obtained with the mordant dyes, even if the dye source were only the local bedstraws, the mordant at the time was likely to have been alum, almost certainly an import. The purple dye found might have been a lichen dye such as the Roman dye orchil.

If the dyed wool was indeed locally produced, the Vindolanda home-dyeing industry in the first century had much of the

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sophistication established in southern Europe. It is possible that some of the techniques and materials had been imported by the Romans themselves, but these had apparently been successfully assimilated by their British subjects.

Acknowledgements

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[9] A. R. Burnett and R. H. Thomson, op. cit. Note 8, p. 854.

[10] R. Hill and D. Richter, op. cit. Note 7.

[11] A. R. Burnett and R. H. Thomson, op. cit. Note 8.

[12] G. W. Taylor, *Dyes on Anglo-Scandinavian Textiles* (to be published).

[13] The colorant in Tyrian Purple, 6,6′-dibromoindigotin, is essentially insoluble in hot pyridine/water, but dissolves in 100% hot pyridine. This solvent would, presumably, extract the colorant from a Tyrian Purple dyed sample but, unfortunately, such a known dyeing was not available for trials.

[14] D. H. S. Richardson, *The Vanishing Lichens* (David and Charles, Newton Abbot, 1975) p. 81.

[15] D. H. S. Richardson, op. cit. Note 14, p. 86.

[16] R. J. Forbes, op. cit. Note 3, p. 100.

[17] A. R. Burnett and R. H. Thomson, op. cit. Note 8.

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